



Deep learning in wheat diseases classification: A systematic review

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Abstract

The main goal of this paper is to review systematically the recent studies that have been published and discussed WD prediction models. The literature analysis is performed based on studies published from January 1997 to February 2021 by following Kitchenham instructions. After inclusion/exclusion and quality assessment criteria screening, a total of 74 studies have been selected. The literature shows that WD is categorized into three (fungal diseases, bacterial diseases, and insect diseases) types. The research analysis shows that most of the work in the literature has been found on wheat stripe rust (60.81%) disease and the most used prediction technique is ANN (13.32%). The results show that accuracy (67%) is the most prominent performance metric and in the year 2020, a maximum number of papers are published on WD. Also, only five studies have used hybrid approaches which are the combination of SVM and NN techniques.

Keywords Deep learning · Neural networks · Support vector machine · Wheat diseases

1 Introduction

Wheat is one of the most common cereal crops [49] in the world. It is in the form of grass type and cultivated in many varieties over the world. Today, technology plays a very important role for mankind and it helps to produce enough food for billions of people worldwide. Although, the world produces a large number of crops to ensure people's food security. Wheat [67] is the second most common crop for food security and subject to millions of tons import in other countries. There are lots of factors that create threats for ensuring food security. Climate change, pollinators, and plant diseases are the potential threats that exist on crops. In each

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country of the world, more than 22.5% of wheat losses [80] have been estimated. About \$86 million [32] of wheat losses in the country are identified due to wheat rust fungicides. Therefore identification, as well as prediction of Wheat diseases (WD) [22] is very important because late identification and prediction lead to monetary loss for farmers and industries. To control the fungicide, spray equipment is used [10] but many researchers pointed out that limited supply of the equipment not only affecting timely control of the outbreak that causes serious health issues but their usage way also give rise to safety issues for the farmers [79] who spray in the fields to control the fungicide. Sometimes, the one-time spray is not sufficient to finish the WD from the core of the farms due to the nature of their land. Therefore, many farms needed spraying more than once. By considering safety and health issues of farmers' identification as well as for treatment of WD is important, effective methods have to be used like drones or robots that help in the detection and spray the antifungal chemicals on the wheat crops without the involvement of the farmers.

One of the important tasks is to detect the diseases of wheat. These diseases can be either identified by plant disease experts or farmers and this process takes a lot of cost and time. Popular computer vision technology [62] has been used to detect WD with the help of wheat images. But still, accurate detection has not been achieved for the larger datasets for a particular season. To improve the detection rate, Deep learning (DL) techniques have been used. Mainly work has been done on three diseases named fungal, bacterial, and insect-based diseases. In the studies, data has been collected either from real-time devices such as the high-quality camera or taken from laboratory labs. Data collection is known as one of the important tasks too. To early detect WD, several imaging methods such as multispectral thermal imaging [27] and Hyperspectral imaging are used. Moreover, in the recent past, it has been seen that new visualization approaches along with modified/improved versions of DL models have been used to achieve better results. To categorize the WD is a hard task, many researchers have also advocated the use of statistical techniques in the past but when compared, it lacks accurate prediction than DL models [71].

All the above-mentioned concepts have inspired me to include a review and results of the DL approaches that have been used since 1997 for wheat disease identification. To perform this study, systematic literature has been carried out between January 1997 and February 2021 by following the Kitchenham [39] suggested guidelines. The last 7 to 8 years have shown a noticeable growth in the identification of WD using DL algorithms. The current study focuses on reviewing the DL models concerning wheat disease detection and prediction. From the results and discussion point of view, eight research questions have been discussed elaborately. The studies which have been selected for review are taken from five major databases namely Scopus, Springer Link, ACM Digital library, ScienceDirect, and IEEE Xplore.

The motivation for work:

- The diseases are an important cause of loss yield in wheat. Due to environmental conditions, approximately 5% of wheat is lost each year [79]. The authors [17] stated that 15–20% of wheat loss is due to fungal, bacterial, and insect diseases each year. Due to the huge amount of wheat losses, several farmers kill themselves. So, the detection of WD is a key aspect concerning financially as well as socially. The detection is done through an experienced evaluator or computer vision techniques. This involves a lot of time. To overcome these issues, computer vision techniques are used for WD detection.
- When assessed the state of the art of WD, there is not a single systematic literature review paper. The current focus of this work is to do a comprehensive and systematic review by searching for different databases and reporting the research gaps for further investigation.

The rest of the article is followed as: Section 2 describes the overview of DL along with their techniques for the detection of WD. Section 3 presents the history, concepts, and general theories regarding WD. Section 4 identifies the research questions; the research method used to analyze the data material for the current study and introduces the selected analytical framework. Section 5 presents the results and discussions of a systematic review. Section 6 concludes and provides recommendations for further research in the area of wheat disease detection.

2 A gentle introduction to visualization techniques

Many developed/modified DL models and various visualization techniques have been implemented to identify and classify the symptoms of WD. The visualization techniques [35, 41] in the past can easily identify the symptoms of WD. The visualization techniques were introduced to understand plant diseases [2, 31] more clearly. The DL models have the greatest advantage of attempting to learn high-level features [8, 20, 52, 97]. This eliminates [71] the need for manual extractions of features in conventional machine learning techniques which are achieved through human input. DL emulates the human brain and works on unstructured data. For the prediction of WD, the dataset is divided into two parts which are classified as:

training data: The training data (image data) is used to train the model.

validate data: It is the compilation of unbiased data used to check the model and used to give the final model an unbiased solution. The test data is different from the training data.

The current study targets DL models for the classification of WD. Several steps are needed to implement the DL models initially from the selection of data sets to visualization mappings are explained in Fig. 1:

The dataset (image data) is divided into training and testing forms of data. After splitting the data, the data is trained using the DL model and their training/validation plots are retrieve to show the significance of the models [16, 76, 89]. The various types of performance matrices are used to classify wheat images. Since, many selected studies use various performance matrices like accuracy, area under the ROC curve to evaluate the model according to their dataset in the results and discussion section.

2.1 Implementation of DL models

Many selected studies use the DL model with visualization techniques which are described in the 2.1.1 section but some of the selected studies don't use the visualization techniques and it is described in section 2.1.2 for the classification of WD. The visualization techniques and their types are described as:

2.1.1 Without visualization techniques

CNN was used to classify the diseases in wheat plants and to demonstrate the model's implication [63, 89]. For the identification of wheat leaf diseases, basic CNN architectures such as Alexnet, Googlenet, Resnet, Overfeat, and VGG architectures were implemented by various authors. Also, most of the authors [43] compare the MbCNN, Alexnet, and VGG-16/19 DL models were used to classify the five WD namely as powdery mildew, bacterial leaf



Fig. 1 Overview of DL Model Implementation

streak, bacterial leaf blight, stripe, and leaf rust. A detailed comparison of DL models was provided through the classification accuracy.

2.1.2 With visualization techniques

The methods used DL models along with their visualization techniques that are implemented to provide a better view of WD. For example, introduced a saliency map for visualizing the wheat plant symptoms and described six different forms of WD named leaf blotch, stripe rust, leaf rust, loose smut, and black chaff using the VGG-FCN-VD16 CNN model. The authors also compared [45] the CNN model with VGG-FCN-S traditional CNN models. This indicates that VGG-FCN-VD16 has high recognition performance than traditional CNN frameworks while using the WDD2017 dataset. The performance was evaluated using the accuracy, recall, F1 score, and overall accuracy of each CNN model [77, 82, 93]. To access the performance of the DL model more accurately, an actual/real background environment should be examined for practical experimentation for wheat plant disease detection.

3 Background

The WD has been found [75] in the year 1922. Figure 2 shows the evolution of WD from the year 1992 to 2004. Wheat is one of the most essential staple grains in the world and it is a calorie source. Furthermore, the first wheat streak mosaic disease was found in 1922 in the Nebraska area [37] of America and the second tan spot wheat disease was identified in 1923 Nisikado area [36] of Japan. Subsequently, the fusarium leaf blotch (1924), stripe rust (1953), loose smut (1953) were determined [50, 72, 73] in Washington, Nebraska, and Luthra respectively. So, it may be assumed that insect-transmitted viruses will have become more important in northern wheat-growing areas. Afterward, modifications in agricultural practices the aphids (1967) [72] wheat disease were identifying in the trestle area. Furthermore, some pathogens had been grown and a new WD septoria leaf spot (1972), flag smut (1981) was found in the Sanderson [84] and Tundu regions [46]. The bacterial spike blight (1987) and bacterial black chaff (1995) are bacterial WD [6, 38] widespread in irrigated areas with abundant rainfall during the growing season. It is most common and is a widely distributed bacterial disease of small grains. Generally, bacterial diseases were identified in the northern regions of India. Other diseases head blight (1993), spot blotch (1994), leaf rust (1997), stem rust (1999), and powdery mildew (2004) WD were observed in Luthra [38], Amritsar [96], and many isolated fields in southern panhandle [5]. While these diseases were occurred mainly [67] in the western part of India state. It has caused damage in the central and eastern parts of the south as well. It is stated that fungal and bacterial pathogens cause high yield loss on wheat on a global scale. Automatic disease detection through real images is a very important and primary problem for sustainable farming and has gained high attention over the past few years.

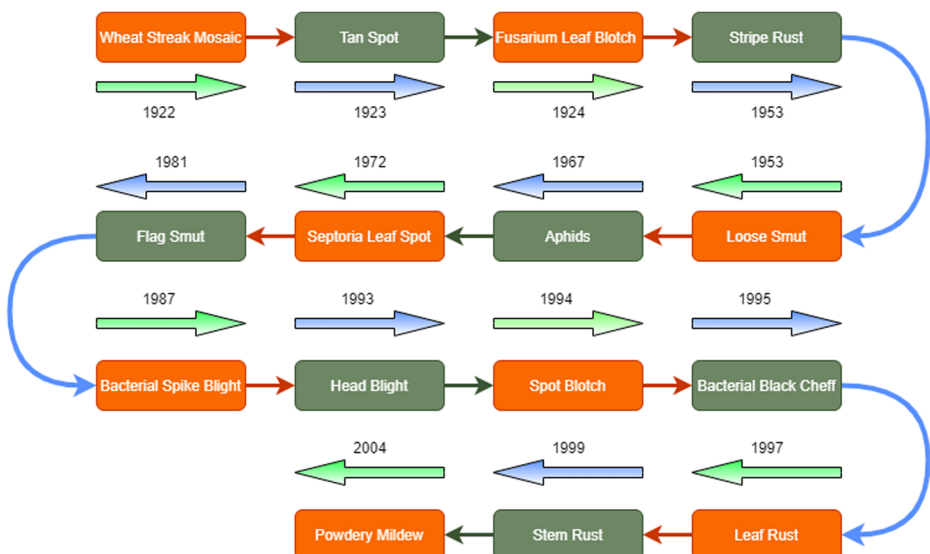


Fig. 2 Evolution of Wheat diseases (1992–2004)

Generally, the WD is classified into three type's namely fungal, bacterial, and insect-based diseases. The types of wheat diseases have been shown in Fig. 3.

The general description of each disease is listed as:

3.1 Fungal diseases

The fungal diseases are caused by fungi [49] and are differ from other plants because they don't have chlorophyll, thus they lose their photosynthetic ability. The fungi are transmitted [32] in many ways: they can be borne by seed or carried by soil or they can spread by wind and water. Infection with fungal pathogens [79] depends on various factors: generally water needed on the host plant surface, host susceptibility and ambient temperature as well as environmental factors. The symptoms of fungi may be identical or distinct. Furthermore, fungal diseases are divided into four categories named rust, leaf spot, spike infection, and virus-based diseases.

The description of each fungal disease along with its subcategories are listed as below:

1. Rust diseases of wheat

There are four rust diseases [79] Disease Management in Wheat that affecting wheat named leaf rust, stripe rust, stem rust, and powdery mildew. Damage to wheat depends on the stage of growth at the time of infection and the overall severity of the rust.

2. Leaf spot diseases

The fungi diseases are the main cause of leaf spots [10]; other leaf spots have been attributed to the herbicide. Also, fungi-induced spot leaf diseases can be confused with bacteria-induced leaf spots and multiple illnesses occur at the same time on the same plants.

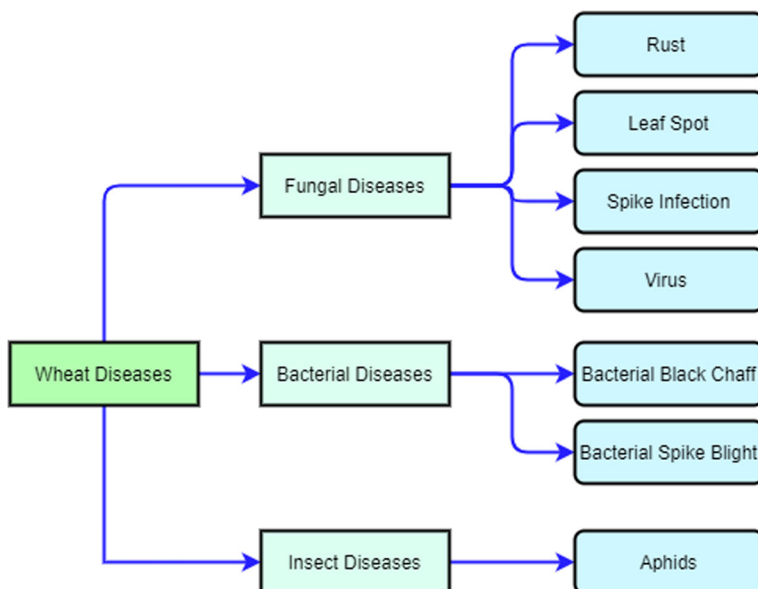


Fig. 3 Types of Wheat diseases

3. Spike infected-based diseases

There are two infected-based diseases [32] that affect the wheat spikes. Before or at another extrusion or at 18 or 28 days later, the fungi were inoculated on winter wheat spikes.

4. Virus-based fungal diseases

The virus-based diseases [79] exist on leaf cells as in flexuous rod forms. The wheat curl mite vector feeds on the growth of young wheat and some grasses. Figure 4 shows the categories along with the subcategories of each fungal disease.

3.2 Bacterial diseases

In bacterial WD, the bacteria can be born from seed [73] and survives in the soil on crop residues tolerating both warm and freezing temperatures. But the moisture is required for bacterial infection and it spread damaged epidermal tissue. This disease is spread through rain or plant touches with another wheat plant and insects. Generally, bacterial diseases are categorized into two forms bacterial black chaff and bacterial spike blight which are described in further sections.

3.3 Insect based diseases

Hundreds of insects are listed worldwide on wheat. Some insect-based diseases are directly [73] linked to the unique farming system used in different areas, while other pests are general herbivores that do not target wheat as a host. Many herbivores are specifically adapted to wheat and WD.

1. Puccinia striiformis f.sp. tritici or stripe rust

It is also called yellow rust disease. These diseases are found when there is a change in temperature from planting to cutting season. The symptoms of this disease with yellow stripe speckles [37] became obvious on leaves at the grain filling stage. The speckles are small,

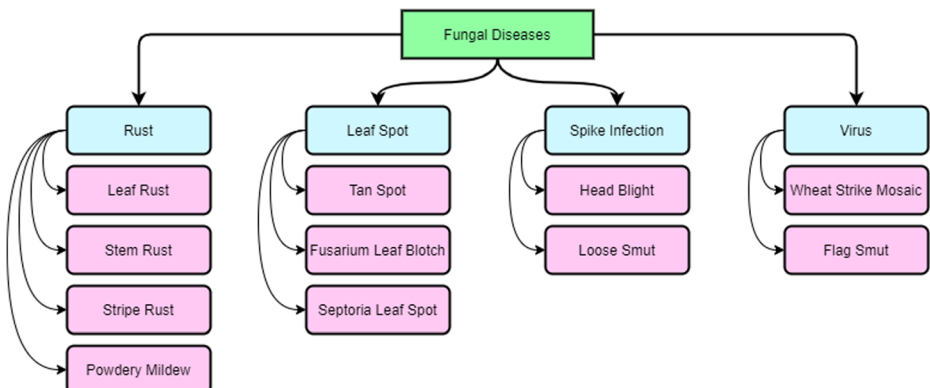


Fig. 4 Categorization of Fungal Diseases

round, and do not cause tears in epidermal tissues at pustule margins. It is observed that the wheat stripe rust field had been started from February to early April. According to a China survey, there is a 29.03% loss [64] of wheat due to yellow rust-infected disease. Figure 5(a) and (b) show the yellow rust diseases on the wheat leaf with pustules and infection respectively.

2. *Puccinia triticina* or leaf rust

This is also known as leaf rust is caused by fungus. The round orange-brown rust appears on the upper leaf surface. This disease [37] occurs when the optimal temperature range between 60 and 70° F and it is observed in the field starting in starting April. Figures 5(c) and (d) show the leaf rust diseases on the wheat leaf with pustules and infection.

3. *Puccinia graminis*f.sp. *tritici* or stem rust

This disease is identified through stem rust. The pustules are oval-shaped [32] with orange-red color in leaves and color is occur on leaves or stems. The wheat stem leaves or root is observed in cutting season [50] and the optimal temperature for these diseases is in the range 75–85° F. Figure 5(e) and (f) shows the stem rust diseases on the wheat stem and leaf with pustules.

4. *Blumeria graminis* f.sp.*graminis* or powdery mildew

This disease is known as powdery mildew. The white powdery mildew spots on leaves and stems. The patches which grow on the plant surface are initially white but due to temperature [50] changes the color turns into dull grey-brown. Figure 5(g) and (h) show the powdery mildew diseases on the wheat stem and wheat leaf with pustules.

5. Bacterial leaf streak or black chaff

Bacterial leaf streak is a pathogen [6] known to infect varieties of wheat and cause damage. This disease losses a huge amount of wheat. This disease is common and can be found in summer and winter wheat varieties. The black chaff is of a yellow cream color and this color is a symptom of the pathogen. The purple and black streaks are found in wheat kernels. Figure 5(i) and (j) shows the bacterial leaf streak and black chaff on wheat plant commonly.

6. Bacterial spike blight or yellow ear rot

This disease is caused by *Pv.syringae* bacteria [6]. The disease progresses between 59 and 77° F temperatures. The bacteria are a poor pathogen and the development of this disease will be curtailed once wheat is exposed to low humidity and high temperature. Figure 5(k) shows the symptoms of bacterial leaf blight on a wheat leaf.

7. *Septoria* leaf spot or *septoria tritici* blotch

This disease is caused due [84] heavy rain and wet weather. The small yellow and brown color pustules on a wheat leaf. The optimal temperature of this disease ranges between 20 and 25° F temperatures. Figure 5(l) shows the symptoms of septoria leaf spot on wheat.

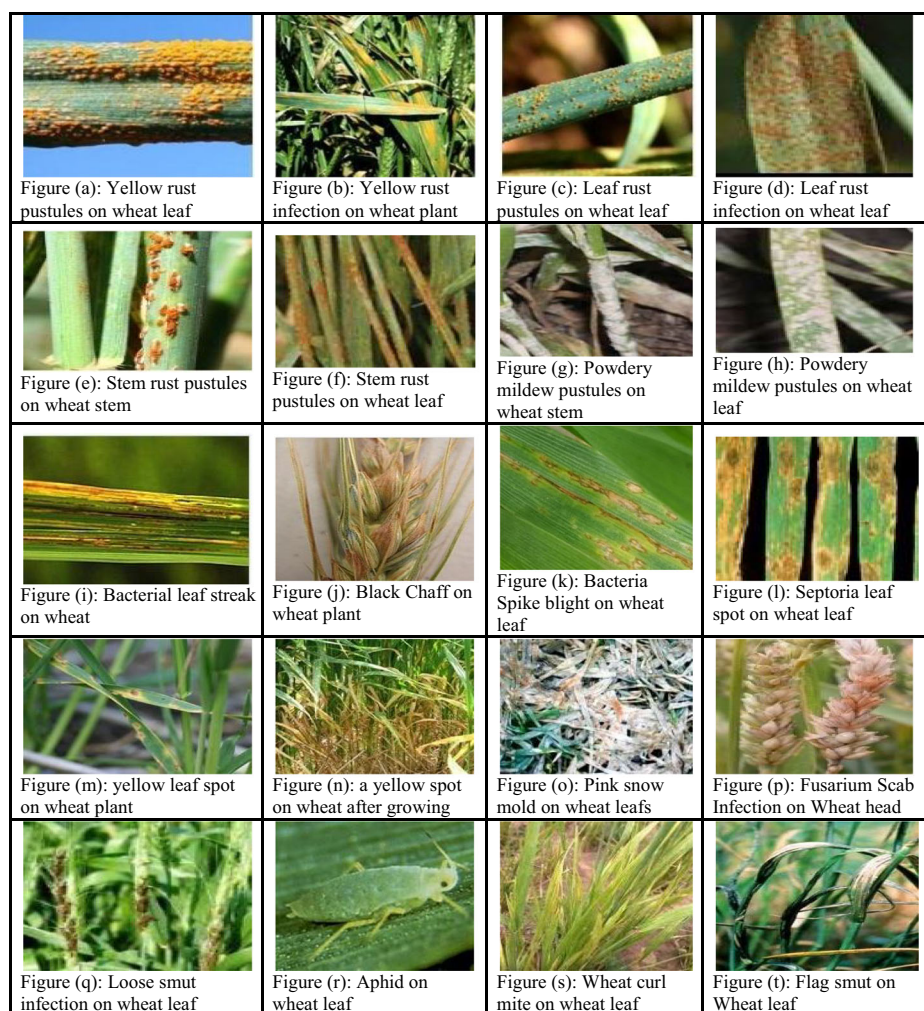


Fig. 5 Wheat Diseases. **a** Yellow rust pustules on wheat leaf. **b** Yellow rust infection on wheat plant. **c** Leaf rust pustules on wheat leaf. **d** Leaf rust infection on wheat leaf. **e** Stem rust pustules on wheat stem. **f** Stem rust pustules on wheat leaf. **g** Powdery mildew pustules on wheat stem. **h** Powdery mildew pustules on wheat leaf. **i** Bacterial leaf streak on wheat. **j** Black Chaff on wheat plant. **k** Bacteria Spike blight on wheat leaf. **l** Septoria leaf spot on wheat leaf. **m** yellow leaf spot on wheat plant. **n** a yellow spot on wheat after growing. **o** Pink snow mold on wheat leaves. **p** Fusarium Scab Infection on Wheat head. **q** Loose smut infection on wheat leaf. **r** Aphid on wheat leaf. **s** Wheat curl mite on wheat leaf. **t** Flag smut on Wheat leaf

8. Yellow leaf spot or *pyrenophora tritici-repentis* or tan spot

The yellow spot is a fungal disease [72] and it can cause the yield losses of wheat up to 30% every year. The yellow-tan oval spots on leaves that are in their center become tan to brown with a yellow edge. Figure 5(m) and (n) shows the symptoms of yellow leaf spot before and after growing on wheat plant jointly.

9. **Fusarium leaf blotch or snow mold or pink mold**

These are fungal diseases [26] and snow molds on wheat are most dramatic and destructive. The symptoms of this disease are found when temperature rises and sunlight increases. Usually, initial infections occur in the fall as well as seedlings grow, and are caused by spores formed from a previous wheat crop on wheat residue. The fungus is of either pink yellow or gray color is found on a wheat leaf. Figure 5(o) shows the pink snow mold on a wheat leaf.

10. **Head blight or scab infection or ear blight**

It is a fungal type of disease and it reduces the quality of grain. The infection is occurring [38] during the flowering stage and the wheat spikes are of pink yellow color. If infection occurs on the stem just below the head, then the entire head can die. Figure 5(p) shows the fusarium scab infection on the wheat head.

11. **Loose smut**

It is a disease borne by the seed. The infection occurs through [61] wind-borne spores during the loose smut flowering. During the cutting season of wheat, the infected heads emerge earlier than normal heads at this time. The infection is favored through cool and moist conditions during the plant's flowering period. Figure 5(q) shows the loose smut on a wheat leaf.

12. **Aphid**

The aphid is an insect-based disease and damages the leaves, flowers of wheat. When the insect is entering [5] into the wheat plant is the main reason for this disease. Wheat is the most important cultivated host and can provide the aphid with suitable habitat for this disease. Figure 5(r) shows the aphid insect on a wheat leaf.

13. **Mites or wheat streak mosaic**

It is a virus-based disease and transmitted through wheat curl mite. The streaked mosaic patterns are observed [36] on wheat leaves and these patterns are symptoms of this disease. During the wheat cutting season if there is humidity temperature (75–80° F) and a hot environment then this disease is more spreading through the curl mite virus. Figure 5(s) shows the wheat curl mite virus on wheat leaves respectively.

14. **Flag smut**

This is a fungal disease and it is caused by the fungus *Urocystis agropyri*. Long [46], black raised streaks on the leaves, sometimes the stems characterize this disease. These black stripes break through the plant tissue, revealing the mass of powdery grey-black spores that can be easily run with the help of a human finger. The flag smut on the wheat leaf is shown in Fig. 5(t).

15. *Helimenthosporium* leaf blight or spot blotch

This is a pathogen-based disease [96] and it is caused due to *cochliobolus sativus* pathogen. When there is a cooler environment then this disease is spreading over the wheat plant. The pathogen is yellow-brown on the wheat plant.

The description of each disease is defined in the above section. All types of WD along with their symptoms are shown in Fig. 5.

The technologies proposed and techniques implemented so far are slightly dependent on computer vision techniques with their scope in the entire paper. The efficiency of computer vision techniques is emerging as the most effective method for diagnosing and predicting pathogens from real-time crop images.

4 Method for conduct the systematic literature review (SLR)

For conducting the WD SLR, the Kitchenman [39] guidelines have been followed which will give the overview as well as a description of the SLR protocol, and it is shown in Fig. 6. The authors have planned to execute the review protocol and define each phase of the review protocol in a better-described form. The phases of the review protocol are following as:

- Identification of research questions
- Construction of search system
- Study selection
- Study quality assessment
- Data Extraction
- Data Synthesis
- Threats to validity

According to the review protocol, the first phase describes the research questions which are associated with the objectives of WD. The second phase of SLR describes the search strategy which aims to analyze those studies which will help in explaining each research question. This phase consists of the search string and literature sources for defining the biased and unbiased search approach. The third phase is based on research questions and finds all consistent studies from selected literature resources. Basically, in this phase, the authors have identified requirements for inclusion and exclusion criteria to drive which studies are to be included and which studies are to be excluded. A criterion for qualitative evaluation calculates the strength and nature of each study. In the data extraction phase, the authors are collecting important information based on research questions and this information is representing in the form of a table. The final phase determines the complete methodologies and synthesizes the obtained data based on research questions.

An important step in SLR is the creation of a review protocol, as it reduces the risk of study bias in SLR. All steps of WD SLR are summarized as well follow as:

4.1 Research questions

The SLR aims at summarizing and clarifying the empirical significance of Wheat disease models based on DL. To this end, the following research questions have been formulated and analyzed:

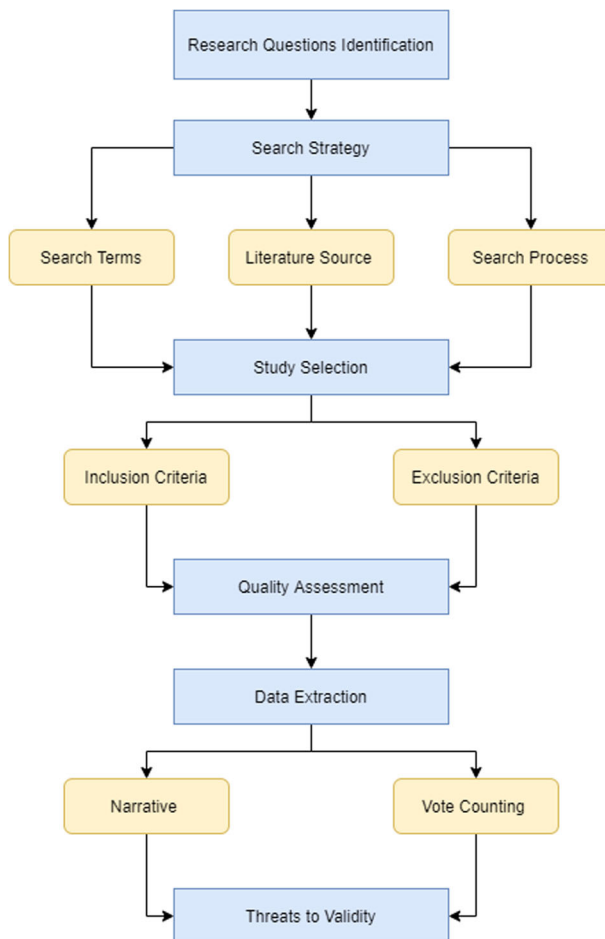


Fig. 6 Phases of Systematic Literature Review (SLR) Protocol

Table 1 describes all research questions along with their motivation.

4.2 Search system

The search system phase consists of search terms, literature sources, and the search process. A description of each phase in the search strategy is following as:

4.2.1 Search terms

The search term is made by using the combination of the below-mentioned steps:

- Find the major terms used from research questions.
- Determine the synonyms for major terms.
- Analyze the keywords in related articles or books.
- The Boolean OR operator used a synonym.

Table 1 Research Questions

RQ#	Research Questions	Motivation
RQ1	What are the various types of WD?	Describes the WD which are identified through deep learning techniques.
RQ2	What image processing techniques are being used?	Find image processing techniques.
RQ3	Which DL models have been used to detect WD?	Analyze the DL models used to detect WD.
RQ4	Is the hybrid approach of DL models have been used to find WD or not?	Identify any hybrid approach used to determine WD.
RQ5	What are the various performance parameters used to evaluate DL models?	Assess the performance of DL models for wheat disease.
RQ6	Which are the data sets most frequently used in WD?	Identify the data sets that are appropriated for WD.
RQ7	Whether the performance of DL techniques is better than statistical techniques?	Investigate the performance of DL techniques over wheat disease statistical techniques.
RQ8	What are the most dominant journals/conference papers analyzing DL models for WD?	Determine the most dominant journals where information was extracted and determine the publication year.
RQ9	What are the various types of wheat images that can be detected using DL models?	Identify the types of images that can be used in the process of DL models.
RQ10	Which are the various types of feature extraction techniques used in wheat disease?	To identify the trends in used feature extraction techniques overall two decades.

e) The AND operator is used with synonym to connect with a major term.

Taking the considerations of WD SLR, the authors have combined Boolean operators OR and AND with includes all synonyms to make the better search string. The authors used disease or diseases and they are picked along with search string. Similarly, the authors have combined neural network techniques or DL or generative adversarial network models on the individual wheat disease.

The resultant search string is:

(Wheat AND (disease OR diseases) AND (“neural networks” OR “neural network” OR “Deep learning” OR “Generative adversarial networks” OR “deep belief networks” OR “restricted Boltzmann machine” OR autoencoders))

After applying the search string on the selected databases, now the secondary search will be applied by looking at the references of the papers that have been selected through the search string.

4.2.2 Literature sources

The literature sources have been focused on searching and selecting the primary studies regarding search string. Based on primary studies, the authors are focused and extract the research paper information using the search string from the following sources:

- IEEE Xplore
- ScienceDirect

- ACM Digital Library
- SpringerLink
- Scopus

The authors are selecting these databases to fetch the studies because these digital libraries are popularly used in the field of wheat disease.

4.2.3 Search process

To extract the related information or particular research paper, a search string has been used primarily on the database (ACM, ScienceDirect, IEEE Xplore, SpringerLink, and SCOPUS) and a secondary search has also been performed. The final selection of studies was based on title, abstract, and keywords. The authors have restricted to start the search from 1993 because the application of DL models was started in the early 1990s. For example, L.J. Francal and S.Panigrahi [18] predict wheat leaf wetness through an Artificial neural network(ANN) in 1997.

A total of 255 studies have been figured out and after inclusion/exclusion criteria, a total of 98 studies has been selected. After rigorously performing a quality assessment, a total of 74 final studies were selected. All the steps are mentioned in Fig. 7.

4.3 Study selection

This phase is also known as the Pilot data selection phase. In this phase, the studies are selecting and searching based on abstract, title, and keywords. Therefore, many candidate

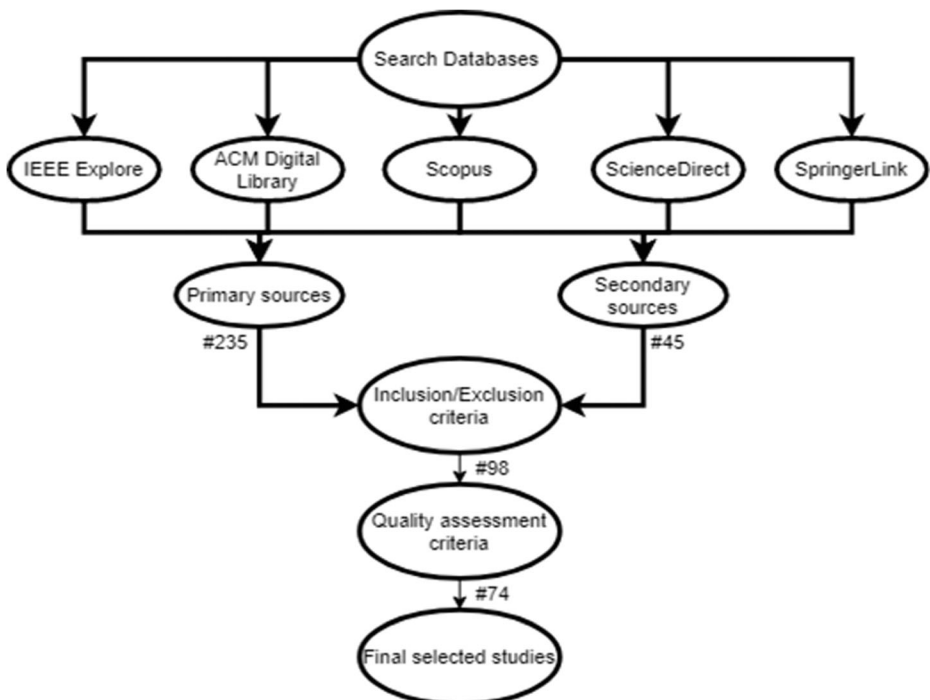


Fig. 7 Data Collection Steps and Refinement

papers give no useful information to answer the research questions. After, applying a filter on candidate papers to identify the relevant papers based on research questions. In this phase, selected studies used the inclusion and exclusion criteria to determine which studies have to be included or exclude during SLR. The inclusion/exclusion criteria have followed as:

Inclusion criteria:

- The authors have looked at the methodology contribute to the studies by preparing two questions:
- Are the authors able to find a clear methodology that answers research questions?
- Is this study being highly related to the research?
- The studies which are published both in journals and conferences, only the journal version of the study was included.
- The studies have mentioned the results/outcomes.

Exclusion criteria:

- Articles that were not relevant to the scope are excluded.
- The studies without empirical investigation are excluded.

After conducting the SLR for WD the authors have several meetings with researchers. After meeting with several researchers, the authors have finished the mutual inclusion/exclusion criteria with a better understanding.

4.4 Quality assessment

The CDR (Center for Reviews and Dissemination) evaluates the SLR [86]. With the help of CDR, the weighing of each paper is measured based on data retrieved in 255 Meta-analysis. The meta-analysis strategy is an important step in the data synthesis phase. To determine the importance and strength of primary studies a consistency questionnaire was developed. The quality questionnaire was built with the suggestions given by Wen et.al [92]. The quality assessments are used for weighing each study.

For each primary question, two independent researchers ranked the quality questions and consulted other researchers in the event of any disagreement. Finally, a definitive decision of inclusion/exclusion criteria for each analysis was made after extensive analyses, debates, and brainstorming sessions with experts and guides. The following questions were used to assess a standard assessment criterion:

- Q1: Are the research aims clearly defined?
- Q2: Is the experiment applied to the dataset appropriately?
- Q3: Is the method for collecting the data specified?
- Q4: Are the DL techniques properly defined?
- Q5: Is the accuracy of models measured?
- Q6: Are the deep learning models compared with other methods (DL models vs. ML, DL vs. statistical methods)?
- Q7: Are the findings and results are clearly stated?
- Q8: Does the study have an agreeable average citation count per year?

For the quality assessment of each study, SLR uses the fuzzy linguistic variables either 0 or 1 for the scoring of these quality assessment questions. The authors think that it's not a better way to weigh them. Taken as an example of quality question "Is the experiment applied to the dataset appropriately?". So, if datasets are found in real-time then the authors will assign value for this question as 1 but what happens if studies have a specific number of datasets? Then the current study approach assigns a score of 0 for the question.

One of the quality assessment question is "Does the study have an agreeable number of average citations count per year (Q8)". In that case, the authors can think that the paper that has publications in the years 2020 and 2021 have low citations as compared to publications in the year 2018 and 2019. The papers of the year 2020–2021 have been included in the current study as they have less citation due to their recent publications. According to their citation, the authors agreed to score the papers as follow:

So, the authors are given a score for each study according to their citation range shown in Table 2. As mentioned above, the authors have measured each quality assessment score for studies published in or after 2019 that only consider seven quality questions. As mentioned above, the questions Q1, Q7 having low score values in quality assessment questions. So, after the inclusion/exclusion filter, this review picked the 74 most important studies.

Since, if there are eight questions about quality evaluation the score for each question is given as:

- 0 No
- 0.5 Partly
- 1 Yes

After several rounds of discussion with researchers about the quality assessment criteria, some of the authors agreed upon the fuzzy linguistic variables to assess the quality of different studies selected. Based on eight quality assessment questions, the total score for every single question is Q1 (1), Q2 (1), Q3 (0.5), Q4 (1.5), Q5 (1.5), Q6 (0.5), Q7 (0), Q8 (1). As a result of giving a score to each quality assessment question, 16 studies have been discarded. In other words, all the studies with variables score "partly and yes" are included in SLR. The nomenclature for reference naming like study name, reference has been inspired from the study [4]. Each study has a different quality assessment score along with its reference that has been shown in Table 4.

4.5 Data extraction

This review manipulated the selected studies by extracting data to obtain useful information. This leads to answering the study questions involved in this analysis. In this phase, the authors have found that some studies used various synonyms for DL models. Certain data could not be

Table 2 Study Scores according to their Citation Range

Sr.No	Citation (range)	Score
1.	0–20	0–0.5
2.	21–40	0.6–1.0
3.	41–60	1.1–1.5
4.	61–80	1.6–2.0
5.	81–100	2.1–2.5
6.	More than 100	2.6–3.0

extracted precisely from the selected studies during this phase. However, the authors could obtain the data of each study indirectly by appropriately processing available data. For example, some studies used “convolutional neural networks” “deep convolutional neural network” some “recurrent neural networks” and some used “artificial neural networks”. These studies can be considered as classes of neural networks. The authors evaluated and assigned a score to each study according to their citation is shown in Table 3. They will give the score to each study according to research questions to describe the credibility. The score ranges from 0 to 10 the minimum score is 0 and the maximum score is 10 for each study.

Each study is evaluated and assigned a score according to their research questions which have been shown in Table 5. Table 6 shows the studies with their related scores. Each study earns one point for each research question. Based on selected studies, there are ten research questions. Each selected study achieves a maximum of ten and minimum zero scores according to the research question answered which is shown in Fig. 8.

4.6 Data synthesis

The main goal of this phase is to classify and group all the evidence relating to a research question from the selected studies. The data collected in this analysis contain both quantitative (e.g., the estimation accuracy of the model) and qualitative data (strength and weakness of DL models). The authors used various methods to synthesize the extracted data related to various types of research questions. The methods for the data synthesis are explained in detail:

- **Narrative synthesis method:** The narrative synthesis refers to an approach [78] in systematic review and synthesis of findings from multiple studies based primarily use the words and text to explain the summarized findings. The authors have evaluated data and use the findings in the table format and various charts. RQ1, RQ2, RQ3, RQ5, RQ6, RQ8, RQ9, RQ10 are evaluated in this manner.
- **Vote counting method:** Vote counting is a simple method for synthesizing evidence [59] from multiple assessments. It simply involves comparing the number of positive studies with the number of negative studies. For RQ4, RQ7 the authors have followed the method of vote counting have followed which allows comparisons of two studies. In this case, the authors have evaluated the accuracy and determine what model outperforms the other models.

4.7 Threats to validity

According to the search string, the relevant studies are selected according to research questions. However, there is a risk that some related studies could have been skipped since some studies

Table 3 Numbers of Papers Count based on their Citation

Sr. No	Citation (range)	Numbers of paper found
1.	0–20	57
2.	21–40	6
3.	41–60	5
4.	61–80	1
5.	81–100	1
6.	More than 100	4

Table 4 Quality Assessment Scores of Each Study

Selected Studies	Reference	Quality Assessment score	Selected Studies	Reference	Quality Assessment score
S1	[22]	Yes	S38	[85]	Partly
S2	[62]	Partly	S39	[55]	Partly
S3	[53]	Yes	S40	[41]	Partly
S4	[43]	Partly	S41	[3]	Yes
S5	[88]	Partly	S42	[27]	Yes
S6	[48]	Yes	S43	[66]	Partly
S7	[11]	Partly	S44	[91]	Partly
S8	[68]	Yes	S45	[19]	Yes
S9	[58]	Yes	S46	[23]	Partly
S10	[9]	Yes	S47	[42]	Yes
S11	[33]	Partly	S48	[30]	Yes
S12	[45]	Partly	S49	[51]	Partly
S13	[18]	Yes	S50	[57]	Yes
S14	[54]	Yes	S51	[34]	Yes
S15	[98]	Partly	S52	[95]	Partly
S16	[60]	Yes	S53	[101]	Yes
S17	[21]	Yes	S54	[40]	Yes
S18	[100]	Partly	S55	[29]	Partly
S19	[99]	Yes	S56	[35]	Partly
S20	[7]	Yes	S57	[83]	Yes
S21	[47]	Yes	S58	[31]	Yes
S22	[2]	Yes	S59	[20]	Yes
S23	[15]	Yes	S60	[52]	Partly
S24	[81]	Partly	S61	[97]	Yes
S25	[102]	Partly	S62	[8]	Yes
S26	[87]	Partly	S63	[1]	Yes
S27	[94]	Partly	S64	[28]	Partly
S28	[69]	Yes	S65	[89]	Yes
S29	[56]	Yes	S66	[76]	Yes
S30	[70]	Yes	S67	[25]	Yes
S31	[74]	Partly	S68	[63]	Yes
S32	[24]	Partly	S69	[16]	Partly
S33	[13]	Yes	S70	[14]	Yes
S34	[64]	Yes	S71	[93]	Yes
S35	[77]	Yes	S72	[12]	Yes
S36	[90]	Partly	S73	[82]	Partly
S37	[44]	Partly	S74	[65]	Yes

do not cover keywords, in the study article and abstract. Although the authors have made every effort to avoid this possibility by referring to the references of selected studies to select all relevant studies there is the possibility that some significant studies have not been selected and it may be assumed as a threat.

5 Result and discussions

The computational model provides a framework within which summarization and extraction of the information from previous studies are performed. Initially, the studies are collected from various search databases. After studies collection, the authors read each study and extract information. This extracted information helps in describing the type of WD, classification

Table 5 Research Questions Answered by Selected Studies

Selected Studies	Research Questions									
	RQ1	RQ2	RQ3	RQ4	RQ5	RQ6	RQ7	RQ8	RQ9	RQ10
S1	✓	✓	✓		✓	✓		✓	✓	✓
S2	✓		✓		✓	✓	✓	✓	✓	
S3	✓		✓			✓		✓	✓	
S4	✓	✓	✓		✓	✓		✓	✓	
S5	✓		✓		✓	✓		✓	✓	✓
S6	✓	✓	✓		✓	✓		✓	✓	
S7	✓				✓	✓		✓		
S8	✓		✓		✓	✓		✓	✓	✓
S9	✓		✓		✓	✓		✓	✓	
S10	✓							✓		
S11	✓		✓					✓		
S12	✓		✓		✓	✓		✓	✓	
S13	✓		✓			✓		✓		
S14	✓		✓		✓	✓	✓	✓	✓	✓
S15	✓	✓			✓	✓		✓		✓
S16	✓		✓		✓	✓		✓	✓	
S17	✓							✓	✓	✓
S18	✓	✓						✓	✓	
S19	✓	✓				✓		✓		
S20	✓		✓		✓	✓		✓	✓	✓
S21	✓		✓		✓	✓		✓	✓	
S22	✓		✓			✓		✓		
S23	✓							✓		
S24	✓	✓		✓	✓	✓		✓	✓	
S25	✓	✓	✓	✓	✓	✓		✓	✓	
S26	✓	✓	✓					✓	✓	
S27	✓	✓	✓		✓			✓	✓	✓
S28	✓	✓	✓					✓		
S29	✓		✓			✓	✓	✓		
S30	✓	✓	✓		✓	✓		✓	✓	
S31	✓	✓	✓		✓	✓		✓	✓	✓
S32	✓		✓		✓	✓		✓	✓	
S33	✓		✓		✓	✓		✓		
S34	✓	✓				✓		✓	✓	
S35	✓	✓	✓		✓	✓		✓	✓	
S36	✓	✓	✓		✓	✓		✓	✓	✓
S37	✓							✓		
S38	✓	✓						✓	✓	✓
S39	✓		✓					✓		
S40	✓	✓	✓					✓	✓	✓
S41	✓			✓				✓		
S42	✓		✓		✓			✓	✓	
S43	✓	✓	✓					✓		✓
S44	✓		✓					✓		
S45	✓	✓	✓		✓	✓		✓	✓	
S46	✓		✓		✓	✓		✓	✓	
S47	✓	✓			✓	✓		✓	✓	✓
S48	✓	✓	✓		✓	✓		✓	✓	✓
S49	✓		✓				✓	✓		
S50	✓	✓						✓	✓	✓
S51	✓	✓	✓	✓	✓	✓		✓	✓	
S52	✓		✓			✓		✓	✓	✓
S53	✓	✓				✓		✓	✓	
S54	✓		✓					✓		
S55	✓	✓	✓		✓	✓	✓	✓		✓
S56	✓		✓		✓	✓		✓	✓	
S57	✓		✓		✓			✓	✓	
S58	✓	✓	✓		✓			✓	✓	
S59	✓		✓		✓	✓		✓	✓	

Table 5 (continued)

Selected Studies	Research Questions									
# Studies	RQ1	RQ2	RQ3	RQ4	RQ5	RQ6	RQ7	RQ8	RQ9	RQ10
S60	✓		✓		✓	✓		✓	✓	
S61	✓		✓		✓	✓		✓	✓	
S62	✓		✓		✓	✓		✓	✓	
S63	✓	✓						✓	✓	
S64	✓	✓		✓	✓			✓	✓	
S65	✓		✓		✓	✓		✓	✓	
S66	✓	✓	✓		✓	✓		✓	✓	
S67	✓		✓		✓	✓		✓	✓	
S68	✓		✓		✓	✓	✓	✓	✓	
S69	✓		✓		✓	✓		✓	✓	
S70	✓		✓		✓			✓	✓	
S71	✓	✓	✓		✓			✓		
S72	✓		✓		✓	✓		✓	✓	
S73	✓		✓		✓	✓		✓	✓	
S74	✓	✓	✓		✓	✓		✓	✓	

technique, dataset availability & type, performance parameters, journal information, and publishing year. The computational model is shown in Fig. 9.

During the discussion, the authors have interpreted the results of the analysis not only within the research questions framework but also in a broader context closely related to the research questions. Some relevant works are also presented to support the findings for justification.

In this section, the authors describe the results for each research question obtained from primary studies.

5.1 RQ1: What are the various types of wheat diseases?

According to selected relevant studies, the WD is classified into three type's namely as fungal, bacterial, and insect-based diseases and it has already been described in section 3.

Most of the authors had been worked on powdery mildew, stem rust, stripe rust, and leaf rust, and yellow rust WD. These diseases are fungus-based and they were predicted or classified with the help of DL techniques. Table 7 shows the comprehensive results of WD.

Figure 10 shows that a maximum of relevant studies have worked on stripe rust and a minimum of relevant studies have been worked on flag smut respectively.

Table 6 The Score of Each Study according to Research Questions Answered

Scores (Range)	Selected Studies
1–2	S10, S23, S37
3–4	S7, S11, S17, S18, S19, S22, S28, S39, S41, S44, S49, S54
5–6	S3, S8, S12, S13, S15, S16, S21, S26, S29, S32, S33, S34, S35, S38, S40, S42, S43, S46, S47, S50, S52, S53, S56, S57, S58, S59, S60, S61, S62, S63, S64, S65, S67, S69, S70, S71, S72, S73
7–8	S1, S2, S4, S5, S6, S8, S14, S20, S24, S25, S27, S30, S31, S36, S45, S48, S51, S55, S66, S68, S74

Table 7 Comprehensive Results of Wheat diseases

Diseases/ Studies	Leaf Rust	Stem Rust	Stripe Rust	Powdery Mildew	Tan Spot	Fusarium Leaf Blotch	Septoria Leaf Spot	Head Blight	Loose Smut	Wheat Streak Mosaic	Flag Smut	Bacterial Black Chaff	Bacterial Spike Blight	Aphids
S1	✓		✓											
S2	✓	✓	✓											
S3			✓									✓		
S4	✓		✓	✓								✓		
S5	✓		✓	✓			✓							
S6	✓		✓	✓		✓								
S7														
S8	✓		✓				✓	✓						
S9	✓	✓			✓									
S10														
S11								✓	✓					
S12	✓	✓	✓	✓		✓						✓	✓	
S13	✓	✓	✓	✓								✓		
S14			✓											
S15			✓											
S16	✓	✓	✓		✓		✓							
S17														✓
S18								✓						✓
S19			✓	✓				✓						
S20														
S21		✓		✓										
S22				✓						✓				
S23				✓										
S24	✓		✓	✓								✓		
S25			✓	✓										
S26	✓	✓	✓											
S27			✓	✓										
S28	✓	✓	✓											
S29			✓	✓										
S30			✓	✓						✓			✓	
S31	✓	✓	✓	✓	✓	✓	✓							
S32	✓	✓	✓							✓				
S33					✓									
S34			✓											

Table 7 (continued)

Diseases/ Studies	Leaf Rust	Stem Rust	Stripe Rust	Powdery Mildew	Tan Spot	Fusarium Leaf Blotch	Septoria Leaf Spot	Head Blight	Loose Smut	Wheat Streak Mosaic	Flag Smut	Bacterial Black Chaff	Bacterial Spike Blight	Aphids
S35		✓	✓											
S36	✓		✓	✓										
S37		✓	✓	✓					✓	✓				
S38	✓		✓	✓		✓	✓							
S39	✓		✓			✓								
S40	✓			✓										
S41	✓	✓	✓											
S42						✓								
S43	✓	✓	✓	✓							✓		✓	
S44			✓											
S45								✓						
S46			✓				✓			✓				
S47			✓	✓										
S48	✓		✓	✓	✓									
S49			✓	✓		✓		✓						
S50	✓		✓	✓				✓						
S51														
S52			✓											
S53	✓		✓											
S54			✓											
S55	✓		✓											
S56				✓										
S57								✓						
S58	✓				✓									
S59	✓	✓												
S60			✓					✓						
S61														
S62			✓			✓								
S63	✓			✓										
S64			✓				✓	✓	✓					
S65		✓	✓											
S66	✓	✓	✓											
S67								✓						
S68			✓											

Table 7 (continued)

Diseases/ Studies	Leaf Rust	Stem Rust	Stripe Rust	Powdery Mildew	Tan Spot	Fusarium Leaf Blotch	Septoria Leaf Spot	Head Blight	Loose Smut	Wheat Streak Mosaic	Flag Smut	Bacterial Black Chaff	Bacterial Spike Blight	Aphids
S69	✓	✓	✓	✓										
S70	✓		✓				✓							
S71			✓	✓										
S72										✓				
S73								✓						
S74								✓						

Table 8 Most Frequently used Image Processing Techniques

Image Processing Techniques	Relevant Studies	Total no. of studies	Percentage (%)
PCA	S1, S4	2	1.48
Segmentation	S6, S18, S24, S28, S30, S31, S34, S36, S40, S43, S45, S47, S50, S51, S53, S58, S64, S71, S74	19	14.06
CWA	S15, S19	2	1.48
Sliding Window Approach	S25	1	0.74
Image Enhancement Techniques	S26, S27, S35, S43, S53, S55, S63	7	5.18
Adjacency Matrix	S38	1	0.74
Spatial Filtering	S48	1	0.74
Smoothing Functions	S40	1	0.74
Data augmentation	S66	1	0.74

Summary of #RQ1 Among all wheat diseases, stripe rust (#40) followed with leaf rust (#31) are prominently used for prediction whereas less amount work has been performed on flag smut disease (#1), Aphid's disease (#2) and Loose smut (#3) prediction.

Table 9 Outline of Image Processing Techniques

Techniques	Benefits	Drawbacks
PCA	- Removes the correlated features from the original dataset [22]. - Minimizes the overfitting problems and enhances visual perception [43]	- Independent variables become more difficult to understand [22]. - Loss of Information [43]
Segmentation	- It's ideal for photographs with more contrast between objects [48, 81, 100]. - The obtained segments are more stable and the detected borders are distinct [69]. - There is no need to follow any complex algorithms to do segmentation [70, 74].	- It is not appropriate for noisy images [64, 69] [90]. - Time and memory are both valuable resources [19, 41, 70, 74]. - The calculation of ridge gradients is difficult [57].
CWA	The signal's amplitude is right as a function of time at time t, and it optimizes the window size [98, 99]	Less efficient and flexible [98, 99].
Sliding window approach	It exactly finds the presence of an object in an image accurately [102].	It is time-consuming and computationally expensive [102].
Image enhancement techniques	- It is most effective for grey scale images [87, 94]. - It's used to modify the image intensity quickly [66, 77].	- It is very difficult to understand the color images in image enhancement techniques [66, 77]. - The original image could not be found [87, 94].
Adjacency matrix	Suitable for high brightness images [85].	It is a more complex technique [85].
Spatial filtering	- Simple in concept - More noise-resistant [30]	Computational expensive [30].
Smoothing functions	Removes the noise from the image easily [41].	Decreases the fine image details [41].
Data augmentation	Improves the prediction accuracy and reduces the labeling cost [76].	More robust, during data generation [76].

Table 10 DL Techniques Found in Relevant Studies

Deep-Learning Techniques	Relevant Studies	Total No. of Studies	Percentage (%)
ANN	S1, S6, S13, S14, S22, S26, S31, S36, S43, S44, S46, S49, S55, S56, S62, S68, S70, S71	18	13.32
RNN	S2	1	0.74
CNN & Its Frameworks	S2, S4, S9, S12, S35, S42, S58, S59, S60, S65, S66, S69, S72	13	9.62
DNN	S2, S51	2	1.48
FCNN	S16, S74	2	1.48
RCNN	S32, S57, S67, S73	4	2.96
RVM	S27	1	0.74
DCNN	S20, S21, S30, S61	4	2.96
Hybrid Approach	S24, S25, S41, S45, S51, S64	6	4.44

5.2 RQ2: What image processing techniques are being used for wheat diseases detection?

Image processing Basically, the image processing means accomplishing some operations on an image to get extract features from an image. Initially, a computer takes input in form images and applies some image processing techniques to extract features from an image in the form of output. Some of the image processing techniques used by selected studies are described as:

- Principal component analysis (PCA)
- Segmentation: Optimizing sharpening filtering algorithm, Linear iterative clustering, K-means clustering, Spectral segmentation
- Continuous wavelet analysis (CWA)
- Sliding window approach
- Image enhancement techniques: Enlargement, Noise Reduction, Rotation, Color image enhancement
- Adjacency matrix
- Spatial filtering
- Smoothing and sharpening functions
- Data augmentation

From the above relevant studies, it can be observed that most of the authors used PCA, segmentation, and image enhancement techniques for processing the image. The quality of an image is easily affected by the environment. The transmission and processing of an image can lead to the blurring and contrast of an image. The PCA is a dimensionality reduction technique and is used to find the patterns in data. The review of image processing techniques used in selected studies is shown in Table 8.

The advantages and disadvantages of each image processing technique have been shown in Table 9.

Summary of #RQ2 The majorly used image processing techniques are segmentation (#19) and image enhancement techniques (#7), whereas the CWA (#2), PCA (#2), sliding window approach (#1), adjacency matrix (#1), spatial filtering (#1), smoothing functions (#1), and data augmentation techniques (#1) are used hardly.

Table 11 Outline of DL Techniques

Classification Techniques	Advantages	Disadvantages	Disease recognition
ANN	- The output of ANNs can be discrete-valued, real-valued [22, 48], or a vector of many real or discrete-valued attributes [2, 18, 54] while the target function can be discrete-valued, real-valued [74, 87], or a vector of numerous real or discrete-valued attributes [66, 90]. - The number of weights in the network and the number of training instances are considered [35], 24], [23].	- The ANN method chooses different parameters during training and has a direct impact on the network's performance [14]. - When the network's error on the sample is lowered to a particular value then it can be considered as incomplete training [66, 90, 93].	Leaf rust [14, 18, 22, 66, 74, 87], Stem rust [18, 66, 74, 87], Stripe rust [8, 18, 22, 23, 29, 48, 54, 63, 66, 74, 87, 90, 91, 93], Powdery mildew [18, 35, 48, 51, 66, 74, 90, 93], Fusarium leaf blotch [8, 48, 51, 74], Septoria leaf spot [14, 23, 48, 74], Tan spot [74], Flag smut [66], Bacterial spike blight [18, 66], Bacterial black chaff [18].
RNN	- In RNN, inputs of any length can be processed. - The model size does not expand as the input size grows [62]. - RNNs can process arbitrary sequences of inputs using their internal memory, whereas feedforward neural networks cannot [62].	Due to RNN complexity, the computation is slow and gradient vanishing problems occur while training [62].	Leaf rust [62], Stem rust [62], Stripe rust [62]
CNN & its frame-works	- High performance [27, 31, 43, 45] - Less time of classification [12, 16, 20, 27, 31, 52, 76, 89] - High classification accuracy [12, 16, 20, 27, 31, 43, 45, 52, 58, 76, 89]	- Computational cost is high [20, 27, 31, 52, 76, 89]. - More time for data training [27, 31, 43, 45, 58]. - Doesn't encode the object location [12, 16, 20, 52, 76, 89].	Leaf rust [16, 20, 31, 43, 45, 58, 62, 76], Stem rust [16, 20, 43, 45, 58, 62, 76, 77, 89], stripe rust [16, 43, 45, 52, 58, 62, 76, 77, 89], Powdery mildew [43, 45], Tan spot [31, 58], Fusarium leaf blotch [27, 45], Septoria leaf spot [58, 89], Loose smut [45, 89], Wheat streak [12], bacterial black chaff [43, 45], Bacterial spike blight [43]
DNN	- High robustness [34, 62] - Parallel computation [34, 62]	- It requires a large amount of data for classification [34, 62] - Extremely expensive [34] - High computation [60]	Leaf rust [62], Stem rust [62], Stripe rust [62], Head blight [34]
FCNN	Reduces the size of convolution layers [19]	High computation [60]	Leaf rust [60], Stem rust [60], Stripe rust [60], Tan spot [60], Septoria leaf spot [60], Head blight [19]
RCNN	It detects the classify each object in an image [24, 25, 83]	- Takes a huge amount of time while training [25, 83] - It is not suitable for real-time images [24, 82]	Leaf rust [24], Stem rust [24], stripe rust [24], Head blight [25, 82, 83], Wheat streak [24]
RVM	It is suitable for probabilistic classification [94]	Predictions are not probabilistic [94]	Stripe rust [94], Powdery mildew [94]
DCNN	- Automatically derived features [47, 70, 82] - Scalable for large amount of image data [7, 70]	In order to perform a classification, a vast amount of data is required [7, 47, 70].	Stem rust [47], Stripe rust [70], powdery mildew [47, 70], Head blight [7, 97], Wheat streak [70], Bacterial spike blight [70]

Table 12 Performance Parameters used in Relevant Studies

Performance parameters	Definition	# Studies
Accuracy	A metric known as accuracy is used to compare the different classification models. This metric is determined when the total number of correct predictions is divided by the total number of predictions. In certain studies, it is also known as precision.	S1, S4, S5, S6, S8, S12, S14, S16, S20, S24, S27, S29, S31, S32, S33, S35, S42, S46, S51, S55, S56, S58, S60, S64, S65, S66, S67, S68, S69, S70, S73
F1 Score/Recall/Precision	It is one of the important metrics used to measure the efficiency of models used in the classification	S25, S57
AUC (area under roc curve)	ROC (receiver operating characteristic curve) is plotted in the x and y-axis respectively. Where the x-axis is plotted as recall and the y-axis is plotted as precision.	S2, S45, S59, S72, S74
Error Guess methods (R ² , RMSE, prediction error, mean validation)	R ² is the proportion of outcome uncertainty described by the predictor variables. RMSE is the root mean square root error which is the square average difference between the real outcome values observed and values expected by the model.	S9, S15, S21, S30, S36, S61, S62, S70

5.3 RQ3: Which dl models have been used to detect the wd?

The DL models that have been used to predict the WD are listed as:

- Neural networks (NN) or Artificial neural network (ANN): General regression neural networks (GRNN), Probabilistic neural networks (PNN), Radial basis function networks (RBF), Linear vector quantization (LVQ)
- Recurrent neural networks (RNN) or Long short-term memory (LSTM)
- Convolutional neural networks: Frameworks of CNN (VGG-16, VGG-19, Alexnet, Densenet, Resnet-50, Resnet-34)
- Deep neural networks (DNN)
- Fully convolutional neural networks (FCNN)
- Region-based convolutional neural networks (RCNN)
- Relevance vector machine (RVM)
- Deep convolutional neural networks (DCNN)

Table 10 shows the number of techniques used for the prediction of WD.

Each DL model has many advantages and disadvantages which have been shown in Table 11.

Summary of #RQ3 Among all DL prediction techniques, ANN (#18) and CNN (#13) are the most prominent techniques whereas RVM (#1), RNN (#1), DNN (#2), and FCNN (#2) are the least recognition methods used in the selected relevant studies.

Table 13 Recognition Rate along with their Performance Parameter of Each Study

# Studies	Performance parameter	Recognition rate (%)
S1	Accuracy	94.29
S2	The area under the ROC curve	DNN (67), Stepwise regression (58)
S4	Accuracy	96.5
S5	Accuracy	SVM (89.23), NN (80.21)
S6	Accuracy	78.1
S8	Accuracy	97
S9	Error guess methods(R^2)	99.3
S12	Accuracy	VGG-FCN-VD16 (97.95), VGG-FCN-S (95.12)
S14	Accuracy	ANN (95.4), QDA (75.6)
S15	Error guess methods(R^2), RMSE	81.0, 110
S16	Error guess methods(R^2)	96
S20	Accuracy	DCNN (97), Random Forest (91)
S21	Error guess methods(R^2), RMSE	80.8, 89.13
S24	Accuracy	93.68
S25	Precision, Recall	85, 86
S27	Accuracy	SVM (66.67), RVM (66.68)
S29	Accuracy	BPNN (81.25), SVM (85.68)
S30	Error guess methods(R^2), RMSE	89, 0.75
S31	Accuracy	98.3
S32	Accuracy	93.4
S33	Accuracy	86
S35	Accuracy	VGG16 (86.53), VGG19 (83.974)
S36	Error guess methods(R^2)	78.9
S42	Accuracy	91.43
S45	The area under the ROC curve	92.5
S46	Accuracy	SVM (95), ANN (53)
S51	Accuracy	74.3
S55	Accuracy	ANN (84.54), Histogram buckets (78)
S56	Accuracy	89.33
S57	F1 Score, Recall, Precision	74.04, 72.10, 76.16
S58	Accuracy	ANN (84), VGG16 (93), GoogleNet (92)
S59	The area under the ROC curve	96.88
S60	Accuracy	97.99
S61	Error guess methods(R^2)	97
S62	Error guess methods(R^2)	72
S64	Accuracy	95
S65	Accuracy	Alexnet (95.10), VGG16 (96.72), Resnet-50 (97.12)
S66	Accuracy	VGG16 (99.06), Resnet-50 (97.08)
S67	Accuracy	77.19
S68	Accuracy	FLDA (63.2), ANN (78.9)
S69	Accuracy	97.88
S70	Error guess methods(R^2)	0.55
S71	Accuracy	94.16
S72	The area under the ROC curve	91
S73	Accuracy	96.81
S74	The area under the ROC curve	92.5

5.4 RQ4: Is the hybrid approach of dl models have been used to find WD or not?

Table 10 shows that 4.44%% of relevant studies used the hybrid approach models when techniques have been widely used compared with other DL models. The relevant study (S24) uses the hybrid approach of CNN with a Linear support vector machine (SVM) known as the CNN-LSVM approach. The CNN-LSVM approach improves the accuracy of the CNN model

Table 14 Selected studies dataset

# Studies	Year	Device	Data size (values)	Target
S1	2012	Camera	185	Leaves
S2	2017	Camera	8554	Leaves
S3	2010	Camera	2500	Leaves
S4	2017	Camera	83,260	Leaves
S5	2017	Camera	120	Leaves
S6	2015	Camera	632	Stripe
S7	2016	Scanner	532 Plots	Leaves
S8	1998	Camera	180	Stem
S9	2018	Camera	8000	Leaves
S12	2017	Camera	50,000	Stripe, leaves
S13	1997	Scanner	273 Plots	Leaves
S14	2004	Camera	5137	Stem
S15	2014	Camera	197	Leaves
S16	2017	Camera	3637	Leaves
S19	2017	Scanner	20 Plots	Stripe, leaf
S20	2020	Camera	96	Leaves
S21	2019	Camera	612	Leaves
S22	2020	Scanner	–	Leaves
S24	2019	Camera	82,344	Leaves
S25	2019	Unmanned Aerial Vehicles	–	Leaves
S29	2014	Camera	–	Leaves
S30	2019	Camera	Five Plots	Leaves
S31	2015	Camera	105	Leaf
S32	2018	Camera	300	Leaves
S33	2000	Scanner	50	Plant
S34	2018	Hyperspectral	113	Leaves
S35	2018	Camera	165	Plant
S36	2014	Camera	120	Stripe, leaf
S45	2019	Camera	1600	Plant
S46	2015	Camera	180	Leaves
S47	2010	Camera	–	Stripe, Leaf
S48	2019	Camera	300	Leaf
S51	2018	Camera	809,200	Leaf
S52	2013	Camera	–	Stripe
S53	2017	Scanner	–	Leaf
S55	2018	Camera	2207	Stripe, leaf
S56	2020	Camera	150	Leaves
S59	2020	CIMMYT	876	Stripe, leaf
S60	2020	Camera	5242	Stripe
S61	2020	Unmanned Aerial Vehicles	1680	Leaves
S62	2020	Camera	630	Stripe, leaf
S65	2020	Camera	9340	Stripe, leaf
S66	2020	Camera	5100	Stripe, leaves
S67	2021	Camera	690	Stripe, leaf
S68	2021	Unmanned Aerial Vehicles	58	Leaves
S69	2020	Camera	1163	Stripe, leaf
S72	2021	Camera	5656	Leaves
S73	2021	Camera	524	Leaves
S74	2019	Camera	1720	Leaves

for the identification of wheat leaf diseases in imbalanced wheat images. The authors used this hybrid approach for reducing the complexity of classification for wheat leaf diseases. While S25 uses the hybrid approach DC-FCNN which is a combination of deep neural networks and a fully convolutional neural network. The relevant study (S41) proposed the hybrid approach

Table 15 Dominant Journals Information

Selected Studies	Publication name	Type	Number	Percentage (%)
S1	International Conference on Natural Computation, IEEE	Conference	1	0.74
S2	Proceedings of IEEE Conference on Computer Vision and Pattern Recognition Workshops	Conference	1	0.74
S3	International Conference on Intelligent Computing and Intelligent Systems, IEEE	Conference	1	0.74
S4, S65	IEEE Access	Journal	2	1.48
S5	International Conference on Intelligent System and Information Management, IEEE	Conference	1	0.74
S6	Proceedings of Third International Symposium on Women in Computing and Informatics	Conference	1	0.74
S7	Proceedings of International Conference on Internet Multimedia Computing and Service	Conference	1	0.74
S8	Cereal Chemistry	Journal	1	0.74
S9	Journal of Science and Food Technology	Journal	1	0.74
S10	Molecular Plant Pathology	Journal	1	0.74
S11, S12, S14, S15, S16, S20, S61	Computers and Electronics in Agriculture	Journal	7	5.18
S13	Agricultural and Forest Meteorology	Journal	1	0.74
S17, S18, S19, S62	Biosystems Engineering	Journal	4	2.96
S21	European Journal of Agronomy	Journal	1	0.74
S22	Information Processing in Agriculture	Journal	1	0.74
S23	Journal of Indian Society of Remote Sensing	Journal	1	0.74
S24	Neural Network World	Journal	1	0.74
S25, S45, S51, S67, S68, S73, S74	Remote Sensing	Journal	7	5.18
S26	International Journal of Emerging Technologies in Engineering Research	Journal	1	0.74
S27	Optical Measurement Technology and Instrumentation	Journal	1	0.74
S28	International Journal of Computer Science Engineering and Information Technology Research	Journal	1	0.74
S29	In International Conference on Computer and Computing Technologies in Agriculture	Conference	1	0.74
S30, S60	Frontiers in Plant Sciences	Journal	2	1.48
S31	Iranian Conference on Electrical Engineering, IEEE	Conference	1	0.74
S32	Plant Methods	Journal	1	0.74
S33	Phytopathology	Journal	1	0.74
S34, S64	Sensors	Journal	2	1.48
S35	Asian Journal of Computer Science and Technology	Journal	1	0.74
S36	Advanced Materials Research	Journal	1	0.74
S37	Annual Review of Phytopathology	Journal	1	0.74
S38	Vibroengineering Procedia	Journal	1	0.74
S39	International Journal of Scientific and Engineering Research	Journal	1	0.74
S40	International Journal of Latest Technology in Engineering, Management & Applied Science	Journal	1	0.74
S41	Journal of Thi-Qar Science	Journal	1	0.74
S42	Applied Sciences	Journal	1	0.74
S43	Proceedings of the International Conference on Data Engineering and Communication Technology	Conference	1	0.74
S44	In International Conference on Computer and Computing Technologies in Agriculture	Conference	1	0.74
S46	Science and Information Conference	Conference	1	0.74
S47	International Congress on Image and Signal Processing	Conference	1	0.74
S48		Conference	1	0.74

Table 15 (continued)

Selected Studies	Publication name	Type	Number	Percentage (%)
S49	In International Conference on Sustainable Technologies for Industry	Journal	1	0.74
S50	IEEE Latin America Transactions	Conference	1	0.74
S52	International Conference on Advanced Mechatronic Systems	Journal	1	0.74
S53	Mathematical & Computer Modeling	Journal	1	0.74
S54	Procedia Computer Science	Conference	1	0.74
S55	International Conference on Computer and Computing Technologies in Agriculture	Conference	1	0.74
S56	International Conference on Next-generation computing	Journal	1	0.74
S57, S58	Plants	Conference	2	1.48
S59	An ASABE Meeting Presentation	Conference	1	0.74
S63	Cognitive Sciences, Genomics, and Bioinformatics, IEEE	Journal	1	0.74
S66	Iraqi Journal of Science	Conference	1	0.74
S69	International conference on intelligent Sustainable systems, IEEE	Conference	1	0.74
S70	International conference on intelligent systems and computer vision, IEEE	Journal	1	0.74
S71	Precision Agriculture	Journal	1	0.74
S72	Sustainable Computing informatics and computing	Journal	1	0.74
	PeerJ computer science	Journal	1	0.74
	Total		74	

Fuzzy logic+NN which is composed of fuzzy logic and ANN for finding the wheat rust diseases in a real environment. For solving classification complexity, the authors proposed the hybrid approach IABC-K-PCNN which is a combination of segmentation technique (K-means clustering) with pulse-based CNN known as improved artificial bee colony K-means-pulse coupled neural network have been found in one relevant study (S45) to identify the fusarium head blight wheat disease. While the selected study (S51) proposed the hybrid approach DCRNN known as deep convolutional recurrent neural networks. Therefore, the selected study (S64) identifies the head blight wheat disease through a hybrid approach known as PSO-SVM and this hybrid approach is a combination of Particle swarm optimization with SVM.

Summary of #RQ4 Only 6 studies have used hybrid models for the prediction of WD. The names of hybrid models are CNN-LSVM, DC-FCNN, Fuzzy logic-NN, IABC-K-PCNN, DCRNN, and PSO-SVM.

5.5 RQ5: What are the various performance parameters used to evaluate the DL models?

Several varieties of parameters are used to evaluate the performance of DL models. These performance parameters are used in relevant studies with the description is shown in Table 12.

Figure 11 shows the count of different performance parameters used by selected studies. The performance parameter along with their recognition/detection/classification rate of each selected study for WD is shown in Table 13.

Table 16 Types of Wheat Images for Wheat disease Classification

# Studies	Wheat images (Types)
S1	RGB
S2	HSI
S3	RGB
S4	RGB
S5	RGB
S6	RGB
S8	RGB
S9	HSI
S12	RGB
S14	RGB
S16	RGB
S17	MSI
S18	HSI
S20	HSI
S21	RGB
S24	RGB
S25	HSI
S26	RGB
S27	RGB
S30	RGB
S31	RGB
S32	RGB
S34	HSI
S35	RGB
S36	RGB
S38	HSI
S40	RGB
S42	HSI
S45	RGB
S46	RGB
S47	RGB
S48	RGB
S50	RGB
S51	HSI
S52	HSI
S53	RGB
S56	HSI
S57	RGB
S58	RGB
S59	RGB
S60	RGB
S61	HSI
S62	RGB
S63	RGB
S64	RGB
S65	RGB
S66	RGB
S67	RGB
S68	HSI
S69	RGB
S70	MSI
S72	RGB
S73	RGB
S74	RGB

Table 17 Features Extraction Techniques

Feature Extraction Techniques	# Studies	Percentage (%)
GLCM	S1, S5, S14, S20, S27, S31, S36, S38, S40, S50, S52, S55	8.88
SGPM	S8, S48	1.48
CCM	S8, S40, S43	2.2
CWT	S15	0.74
Gaussian Model	S17	0.74
Gabor Filter	S43, S47	1.48

The different model achieves different accuracy during recognition. The prediction techniques have been categorized into three types namely statistical, DL, and hybrid techniques which have been shown in Fig. 12 along with their highest accuracy.

Summary of #RQ5 Among all performance parameters of WD prediction, accuracy as a performance metric is found to be most prominently used in both DL or Non-DL based techniques of prediction.

5.6 RQ6: What are the data sets most frequently used in WD?

Various types of datasets have been used for the classification of WD. These datasets are either publicly or privately available to the scientific community. The publically datasets are available openly but the private dataset is used by only a single researcher. There is no publically [33, 43–45, 71] dataset available for WD. Each researcher took the real-time dataset in the form of images with the help of either camera or sensors. These images are used for either classification or prediction of the WD. Different type of datasets used by various researchers along with various regions is shown in Table 14 as:

From the analysis of Table 14, one point to be noted that there is no publically dataset has been used for wheat disease prediction.

Summary of #RQ6 Most of the datasets used in the selected set of studies are private (not publically accessible) and the study with the largest dataset is having 83,260 wheat RGB images.

Table 18 Summary Points of Wheat disease SLR

Subject matter	SLR Summary
Wheat Diseases	Most frequently: Stripe rust Less frequently: Flag smut
Image processing techniques	Segmentation and other techniques
Topmost used DL models	NN, CNN, and Hybrid approaches
Top 3 widely used performance parameters	Accuracy, Error guesses methods, and AUC
Topmost journals	Computers and electronics in agriculture, Remote sensing, Biosystems engineering
Topmost countries of WD research	China, USA, India
Topmost widely used feature extraction techniques	GLCM, CCM, and SGPM

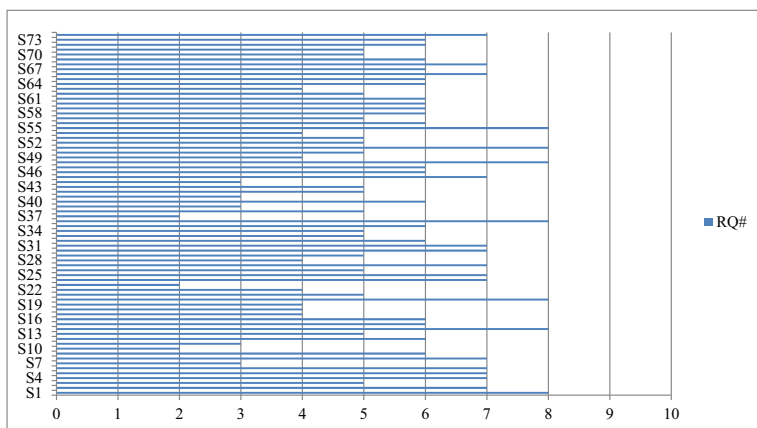


Fig. 8 Overall Scores of Selected Studies

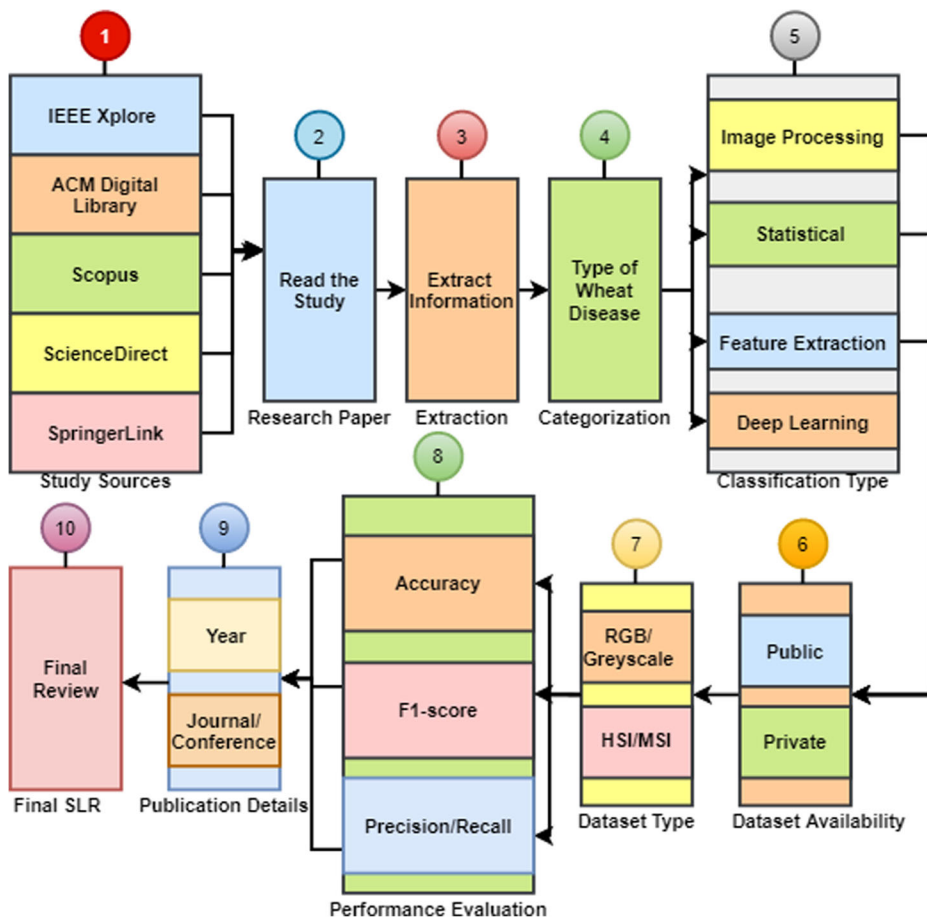


Fig. 9 Computational Model for Data Collection

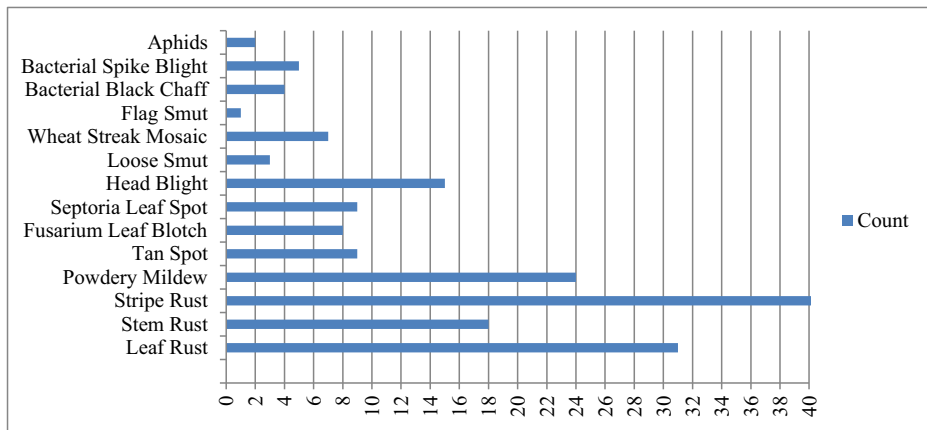


Fig. 10 Count of Different Wheat diseases

5.7 RQ7: Whether the performance of DL techniques is better than statistical techniques?

In this section, the authors will summarize the statistical techniques used to classify the WD. Some of the relevant studies S2, S13, S14, S29, S49, S55, and S68 compare DL models with statistical techniques. During the comparison, each selected study shows that DL techniques achieve high performance as compared to statistical techniques for WD classification as mentioned in Table 11. The statistical techniques are listed as:

- Histogram buckets (HB)
- Stepwise regression (SR)
- Quadratic discriminant analysis (QDA)
- Fisher linear discriminant analysis (FLDA)
- Equilateral Normalization (EN)

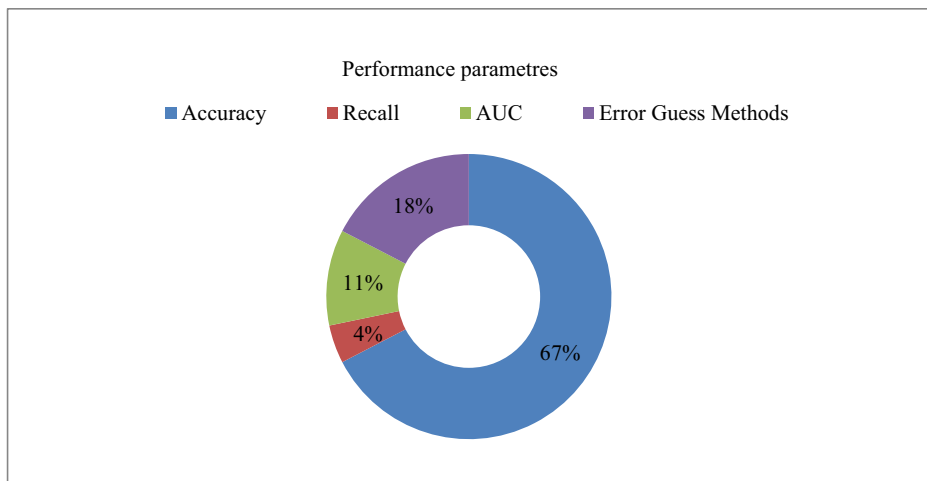


Fig. 11 Performance Parameters Used by Each Selected Study

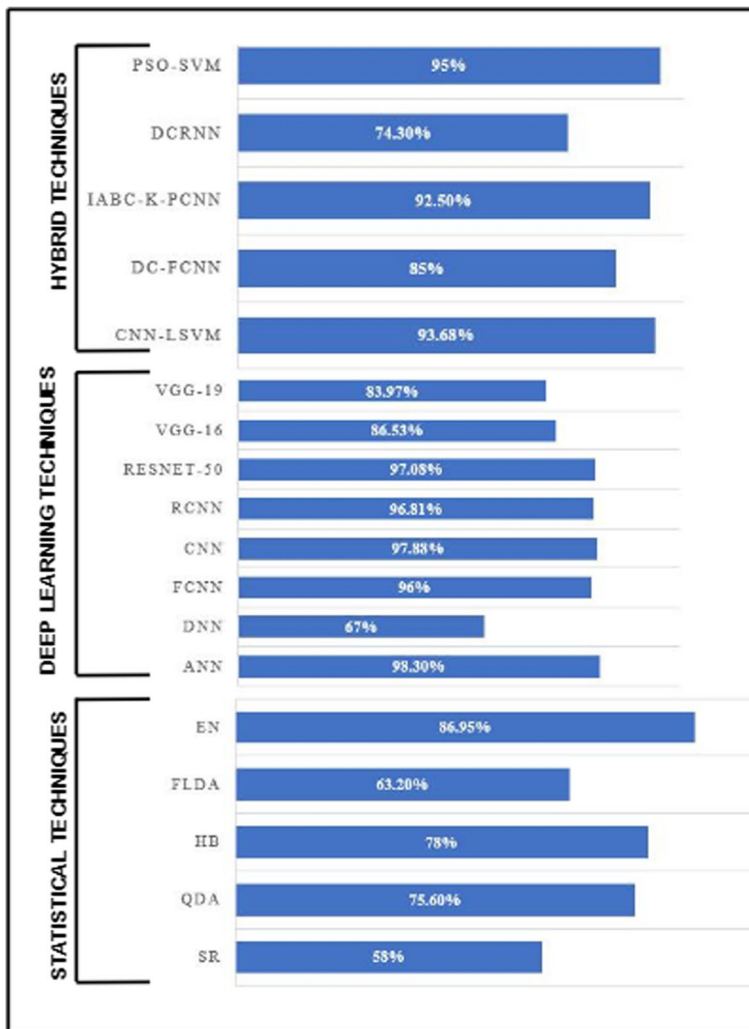


Fig. 12 Accuracy of Different Models

During the comparison, the relevant studies show that DL models are better than statistical techniques.

Summary of #RQ7 The comparison results show that DL techniques outperform the Statistical techniques in terms of accurate prediction. Among all studies, only 7 prominent studies have used different statistical techniques for prediction.

5.8 RQ8: What is the most dominant journal/conference papers analyzing the DL models for WD?

The descriptions of the publication in top journals and conferences are listed in Table 15.

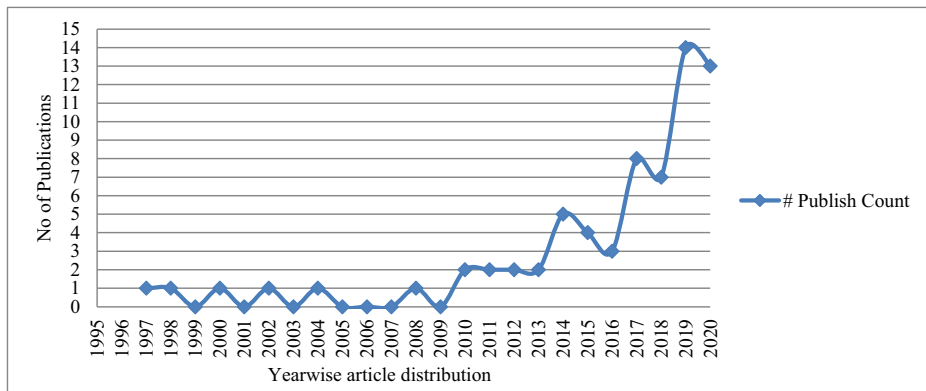


Fig. 13 Year-wise Studies Distribution

Table 15 shows the most dominant journals that are selected for WD SLR. It can be observed that 65% of studies have been published in journals while the remaining (35%) in conference proceedings. Computers and electronics in agriculture, remote sensing, and Biosystems engineering are the most dominant journals used in the WD SLR.

5.8.1 Publication over the year

Figure 13 demonstrates the selected studies are distributed from 1997 to 2021. Figure 13 indicates that only 51 studies out of 74 investigated the use of DL techniques for WD after 2015. Several studies were conducted after 2005. The number of studies found in the years 2016, 2017, 2018, 2019 and 2020 was 3, 8, 7, 14, and 13 respectively. But in the year 2021, only 6 studies have been found till February now.

Summary of #RQ8 From the beginning of 1994 to the year 2018, the research activities are infrequent in number. After that, the period considered from 2019 to the present time, the wheat disease research has been increased. Among all journals and conferences till 2021, the most dominant journal is computer and electronics in agriculture for WD prediction.

5.9 RQ9: What are the various types of wheat images that can be detected using DL models?

Various types of wheat images can be used for the detection of WD. For preprocessing, some of the selected studies have been used different types of color images. The most frequent color images are used by different selected studies as follows:

- Red, Green, and Blue (RGB)
- Hyperspectral imaging (HSI)
- Multispectral Images (MSI)

The UAV is an image acquisition platform. Therefore, a wide variety of sensors can be used with UAVs for image acquisition. Consequently, it is possible to acquire or record RGB,

multispectral, hyperspectral images depending on the sensor used by the UAV. Most of the selected studies have used RGB images. Also, one of the selected studies S38 uses different combinations of red with green and red with blue, etc. Some of the selected studies use Hyperspectral and UAV images for obtaining the spectrum of each pixel in the image. The HSI imaging system works in either visible or short-wave infrared (VNIR) or long-wave near-infrared (SWIR) for image spectroscopy. Therefore, each selected study uses different types of wheat images that have been used for WD classification and it is shown in Table 16.

Summary of #RQ9 Among all color spaces, three-color spaces images RGB (#39), HSI (#13), MSI (#2) are mostly used for WD detection.

5.10 RQ10: Which are the various types of feature extraction techniques used in wheat

5.10.1 Diseases?

Features extraction is an important component of SLR for WD. Several selected studies extract the features from an image through conventional features extraction techniques. Various types of feature extraction techniques have been used in the studies selected by SLR. When the authors applying some feature extraction techniques to the image then color, shape, and textures features are extracted. But some of the selected studies use the DL models for feature extraction because they can extract the features from an image automatically. Here, some of the feature extraction techniques used by selected studies have been shown in Table 17. The percentage of each feature extraction technique used by selected studies has been shown in Fig. 14.

- Grey-Level Co-occurrence Matrix (GLCM)
- Spatial Grey-level dependent Method (SGPM)

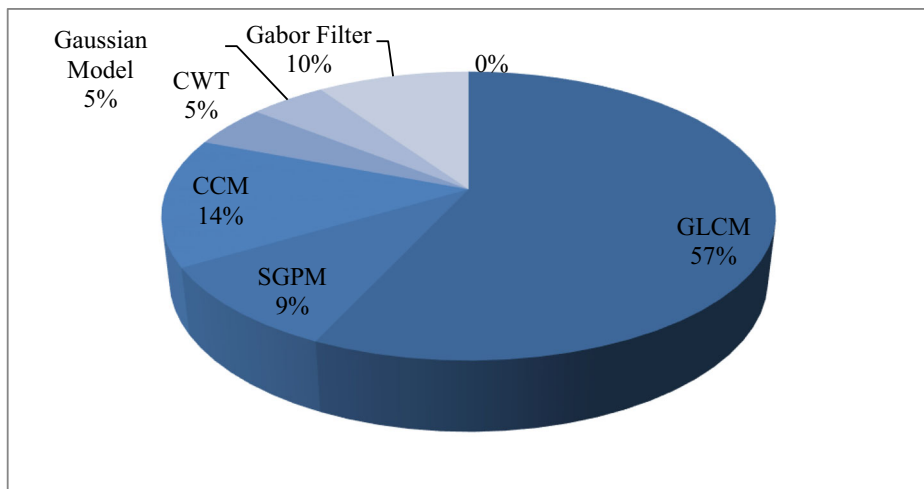


Fig. 14 Extensive Feature Extraction Techniques

- Color Co-occurrences Matrices (CCM)
- Continuous Wavelet Transform (CWT)
- Gaussian Model
- Gabor Filter

Summary of #RQ10 The most prominent used feature extraction techniques are GLCM (#12) and CCM (#3), whereas the SGPM (#2), CWT (#1), Gaussian model (#1), and Gabor filter (#2) are hardly used.

6 Summary and findings

The authors are representing the outcomes of a WD SLR which is related to wheat disease detection using DL approaches. Only 74 studies have been selected that are extracted from various digital libraries to include the IEEE Explorer, ScienceDirect, ACM Digital Library, Springer, and Scopus databases between the span of January 1997 to February 2021. The WD SLR shows that most of the selected studies are published in the year 2019 and only one of the studies has been published in the year 2021 till February. This section addresses the principal results of systematic reviews, strengths, and weaknesses of the evidence. Implications for researchers and some practitioners are summarized after that accompanied by the limitations of the study:

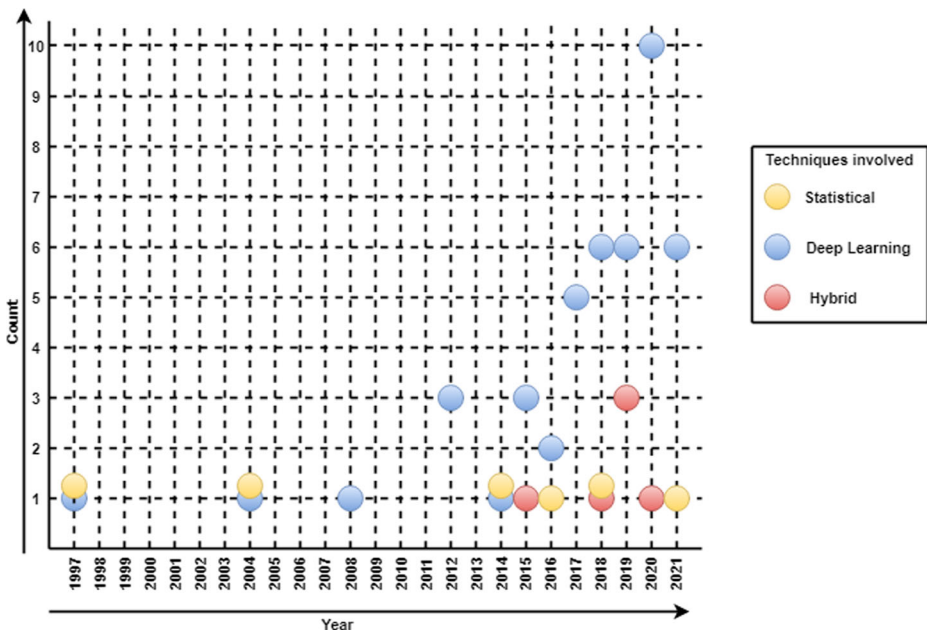


Fig. 15 Year-wise development model

- The WD is divided into three categories. It has been noticed that most researchers are working on stripe rust fungal disease and virus-based fungal diseases less work has been found.
- Publically availability of the WD dataset is not available for classification.

Table 18 shows the summary points of SLR for WD.

Ultimately, in the final, the authors are wishing to emphasize that the primary aim of WD SLR was to review all possible studies on DL models for WD detection between 1997 and 2021 till February. The WD SLR has considered the studies published in the period selected after to determine the datasets, DL models used for the detection of WD.

6.1 Year-wise development model

Different models have been used for WD detection in different years. The year-wise development model (from 1997 to 2021) for the WD detection is shown in Fig. 15.

6.2 Suggestions for researchers

From the above considerations, the authors would like to highlight some findings where the researchers might exploit in the future to carry out DL models for wheat disease detection. The summary of each consideration from the selected studies is followed as:

Most of the authors have been worked in wheat rust diseases but there is less work in bacterial and other fungal WD. Also, various types of DL models are used for the prediction of WD but ANN is most widely used. Therefore, the authors encourage researchers to use a wide range of DL architectures such as GoogleNet, Resnet, SegNet, AlexNet, VGGNet, ResNeXt, Squeezenet, GAN, LetNet, ZFNet, and hybrid approaches for WD prediction. Also, DL applications help to find the wheat quantification, lodging, severity levels, grain variety in terms of wheat products. The accuracy of wheat disease prediction improves when choosing the most important features of datasets. Hence, each of the selected studies uses the wheat disease dataset privately. However, from the above conclusion publically available WD datasets are not found. The authors are given suggestions for researchers to make a wheat dataset publically so that most of the wheat disease researchers can easily use these datasets for better prediction.

7 Conclusion

In this paper, the results of wheat diseases detection SLR using DL models have been studied. The number of selected studies is 74, which the authors reviewed and analyzed to extract important information for users. The studies are selected between the years 1997 to 2021; the authors assume a tremendous amount of research has been done over the last seven years on the detection of WD based on DL models. In particular, the authors have considered ten research questions where all the information is gathered from selected studies. From the observation point of view, most of the selected studies were working on stripe rust fungal wheat disease in China country respectively. As for the findings and comparison, the authors observed that most NN and ANN DL techniques are used in the detection of WD frequently. But during the detection of WD, the ANN achieves more accuracy than NN models. Most of the authors are working on DL models, but only four studies have been found that used the hybrid approach for Wheat Disease Detection.

About the future work, the authors intend to perform the hybrid approach that combines machine learning with the DL model for the detection of WD. The literature shows [43, 66, 101] that the Generative Adversarial Network (GAN) has not been used previously for data augmentation in the case of wheat crops.

Thus, the authors need to exploit new deep learning techniques for improving the detection accuracy on WD is identified, making this area of study potentially active and complex for both researchers and practitioners.

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